Xiang LIU

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Education

- **The Hong Kong University of Science and Technology (Guangzhou)** — **Guangzhou, CHINA**
  *PhD in DSA Thrust, supervised by **Professor Xiaowen Chu** and **Professor Xuming HU*** — *Sep 2023 - Aug 2027*
- **New York University** — **NY, USA**
  *Visiting PhD in CDS, supervised by **Professor Eunsol Choi*** — *July 2025 - Jan 2026*
- **The University of Hong Kong** — **Hong Kong SAR, CHINA**
  Master of Science(MSc) in Computer Science — *Sep 2022 - Aug 2023*
- **George Mason University** — **VA, USA**
  Bachelor of Science in Computer Science; GPA: 3.71/4.0 — *Aug 2018 - Aug 2022*
  Honors/Awards: *Dean's List (2018-2020)*

Selected Research

First / co-first author: 10 papers (ICML ×2 · ICLR ×2 · NeurIPS ×3 · EMNLP Findings ×2 · Workshop ×1)

- **SIM: Benchmarking and Preserving High-Density Reasoning in KV Cache Compression** — **ICML 2026**
  First Author · *KV Cache, Reasoning, Compression, Benchmark*
- **MDN: Parallelizing Stepwise Momentum for Delta Linear Attention** — **ICML 2026**
  Co-first Author · *Linear Attention, Delta Rule, Parallelization, Momentum*
- **δ-mem: Efficient Online Memory for Large Language Models** — **NeurIPS 2026**
  Author · *Online Memory, Delta Rule, Agents, Associative Memory*
- **NestedKV: Nested Memory Routing for Long-Context KV Cache Compression** — **EMNLP 2026**
  Author · *KV Cache, Long-Context, Training-free Compression*
- **CondenseVLM: Visual Token Condensation for Efficient Multimodal Reasoning** — **EMNLP 2026**
  Author · *VLM, Visual Tokens, Multimodal Reasoning, Token Condensation*
- **EVOQUANT: Self-Evolving Verifier-Guided Strategy Optimization** — **FinNLP 2026**
  Author · *Quantitative Trading, LLM Agents, Strategy Optimization*
- **DiffAdapt: Difficulty-Adaptive Reasoning for Token-Efficient LLM Inference** — **ICLR 2026**
  First Author · *Adaptive Inference, Reasoning LLM, Token Efficiency, Entropy Analysis*
- **Reasoning Language Model Inference Serving Unveiled: An Empirical Study** — **ICLR 2026**
  Co-first Author · *LLM Serving, Reasoning System, Empirical Study, Memory Optimization*
- **ChunkKV: Semantic-preserving KV cache compression for efficient long-context LLM** — **NeurIPS 2025**
  First Author · *KV Cache, Inference, GPU Memory, Long-Context*
- **Perovskite-LLM: Knowledge-Enhanced LLMs for Solar Cell Research** — **EMNLP Findings 2025**
  Co-first Author · *Domain-Specific LLM, Knowledge Augmentation, Materials Science*
- **LISA: Layerwise Importance Sampling for Memory-Efficient LLM Fine-Tuning** — **NeurIPS 2024**
  Co-first Author · *Parameter Efficient Fine-tuning, Layer-wise Optimization*
- **Should We Really Edit Language Models? On the Evaluation of Edited LLMs** — **NeurIPS 2024**
  Co-first Author · *Model Editing, Benchmark, Model Robustness*
- **LongGenBench: Long-context Generation Benchmark** — **EMNLP Findings 2024**
  First Author · *Long-context Generation, Long-context LLMs, Logical Coherence*

Experience

- **UniPat** — **Remote**
  Algorithm Research Intern — *July 2026 - Present*
  - Algorithm research internship (ongoing).
- **Mind Lab / Macaron AI** — **Remote**
  Research Intern — *April 2026 - July 2026*
  - Post-trained **GLM-5.1** and **Qwen3.6-35B** for tool use and code generation, lifting agentic performance on VitaBench, PinchBench, Tau3Bench, SWE-Bench, and Terminal2Bench.
  - Strengthened the model harness for long-horizon, multi-turn tool calling and code execution, improving robustness over extended trajectories.
  - Co-authored **δ-mem** (NeurIPS 2026), **MinT**, and the **Macaron-V1** technical report (arXiv:2608.09819).

- **NYU Center for Data Science** **NY, USA**
  Visiting Student ***Supervisor: Prof. Eunsol Choi*** *July 2025 - Jan 2026*
  - Proposed **DiffAdapt**, a novel algorithm for efficient adaptive reasoning for LLMs (Accepted to ICLR 2026).
  - Conducted comprehensive entropy analysis on reasoning traces to identify "overthinking" phenomena.
- **HKUST Statistics and Machine Learning Research Group** **HK,CHINA**
  Research Intern ***Supervisor: Prof. Tong Zhang*** *Dec 2022 - Aug 2023*
  - Proposed **LISA**, a novel algorithm for efficient fine-tuning of LLMs, accepted at *NeurIPS 2024*.
  - Contributed to the LMFlow framework, enabling personalized LLM fine-tuning and deployment (LLaMA, Bloom, Vicuna).
  - Conducted research on Chain-of-Thought (CoT) methods to enhance logical reasoning in LLMs.
- **Baidu Research Cognitive Computing Lab** **Beijing,CHINA**
  Research Intern *Dec 2021 - June 2022*
  - Worked on dependency parsing using Open Information Annotation (OIA) to convert sentences into DAGs.

Academic Service

- **Area Chair** **NAACL 2027**
  NAACL 2027 Area Chair (ARR Oct)
- **Awards**
  ICML 2026 Gold Reviewer

Competitions

- **Kaggle | Feedback Prize - Predicting Effective Arguments Competition** **Jun 2022 - Aug 2022**
  Team Leader | Silver Medal (Top 2%)
  - Developed baseline code, designed data preprocessing strategies, and optimized model structures.
  - Implemented token classification replacing sequence classification, boosting rank and efficiency.

Professional Skills

- **Languages**: English, Mandarin
- **Computing Skills**: PyTorch, Git, Linux